

# SIMATS ENGINEERING

## CO3- AT2 - Design Task - Medium

**COURSE NAME:** Advance Wireless Communication Systems /

**5G / 6G Communication Systems**

**COURSE CODE:** ECA56

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### COURSE OUTCOME COVERED

**CO: 3** - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

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Dr.

**QUESTIONS: 1 (1 X 100 = 100)**

**Total Marks: 100 Duration: 3hrs**

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)
19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)

27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

***Code of Conduct:***

***I P. Reddy Suhail(192312671L)*** (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: Suhail***

**RUBRICS FOR CALCULATION:**

| <b>Design Task</b>          |                  |                                                                                   |                                                      |                                               |                                              |
|-----------------------------|------------------|-----------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------|
| <b>Criteria</b>             | <b>Weightage</b> | <b>Level 4<br/>(Exemplary)</b>                                                    | <b>Level 3<br/>(Proficient)</b>                      | <b>Level 2<br/>(Developing)</b>               | <b>Level 1<br/>(Beginning)</b>               |
| Understanding of Design     | 20%              | Demonstrates deep understanding of design objectives, constraints, and context    | Shows good understanding with minor gaps             | Partial understanding; misses key aspects     | Lacks understanding of the design context    |
| Evaluation of Design        | 25%              | Provides in-depth critique identifying strengths, weaknesses, and design gaps     | Identifies key issues with reasonable analysis       | Limited critique; mostly descriptive feedback | No meaningful critique provided              |
| Justification               | 20%              | Supports critique with strong reasoning, evidence, and relevant principles        | Provides justification with some supporting evidence | Weak or unclear justification                 | No justification for feedback                |
| Suggestions for Improvement | 20%              | Offers innovative, practical, and well-justified improvement suggestions          | Provides useful suggestions with minor gaps          | Suggestions are limited or partially relevant | No constructive suggestions provided         |
| Communication               | 15%              | Communicates feedback clearly, professionally, and engages actively in discussion | Communicates well with minor issues                  | Communication lacks clarity or engagement     | Poor communication and minimal participation |

### Question 25

## Design an NB-IoT Architecture for Smart Utility Monitoring

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### 1. Understanding of Design — Objectives, Constraints and Context

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#### 1.1 Problem Context

Smart utility monitoring requires periodic, reliable collection of consumption/flow data from very large numbers of electricity, water and gas meters distributed across a city or region. A large fraction of these meters sit in deep-coverage locations — basements, underground pits, utility cabinets and dense-urban structures — where conventional cellular coverage is weak. Meters are typically static, battery-powered (mains supply is often unavailable at the meter itself), generate very small data volumes per report, and are expected to operate unattended for eight to ten or more years without a battery change.

#### 1.2 Design Objectives

- **Extended coverage:** reach deep-indoor and underground meter locations with a target Maximum Coupling Loss (MCL) of up to 164 dB.
- **Massive connectivity:** support tens of thousands of devices per cell for city-scale Advanced Metering Infrastructure (AMI) rollout.
- **Ultra-low power:** achieve a battery life of ten or more years using ultra-low duty-cycle operation.
- **Low cost:** use inexpensive modules and licensed spectrum with minimal incremental infrastructure investment.
- **Security:** provide 3GPP-grade mutual authentication and ciphering appropriate for billing-grade utility data.

#### 1.3 Design Constraints

- **Bandwidth:** only 180 kHz of spectrum is available per NB-IoT carrier, which fundamentally limits achievable throughput.
- **Regulatory:** the system must operate within an existing Mobile Network Operator's licensed spectrum and infrastructure.
- **Latency vs. coverage:** coverage extension through repetition trades off directly against latency and cell capacity.
- **Interoperability:** the architecture must interoperate with existing utility back-end systems such as MDMS, SCADA and billing/ERP.
- **Scalability:** the design must scale from a pilot deployment to a full city/regional rollout without architectural rework.

### 2. Proposed NB-IoT System Architecture

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The architecture is organised into five functional layers, from the metering device up to the utility application layer, as shown in Figure 1.

## NB-IoT Architecture for Smart Utility Monitoring

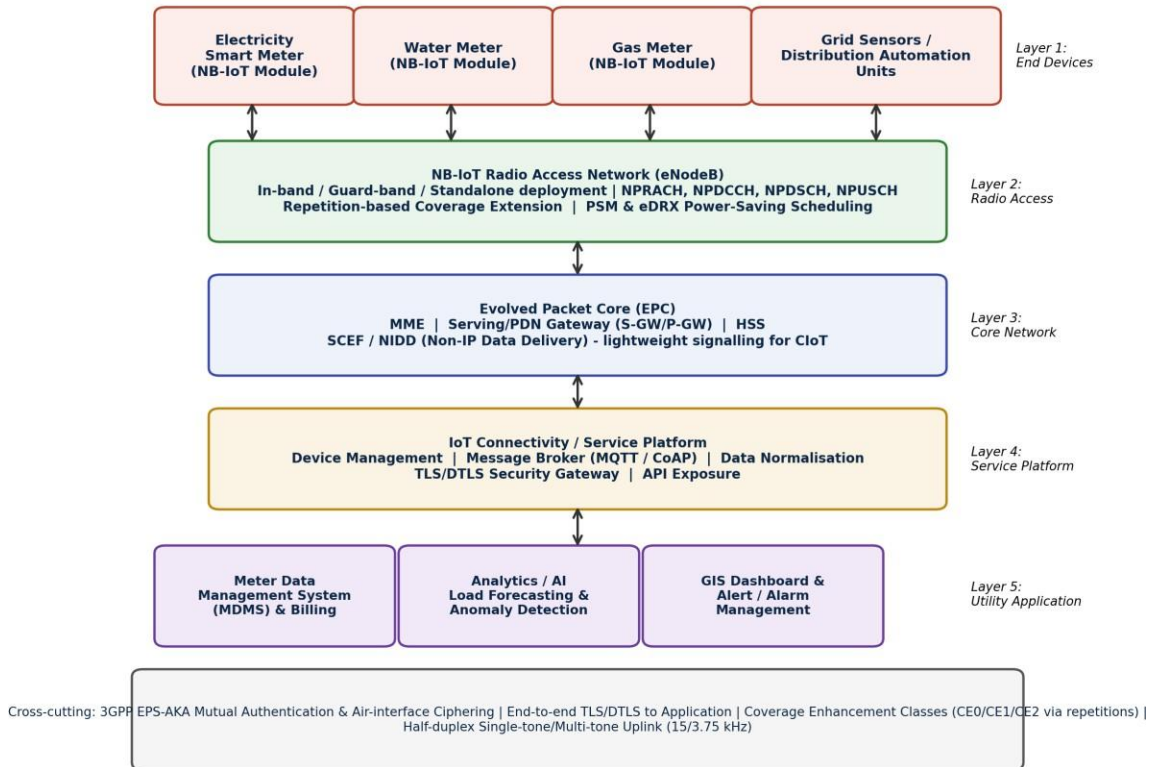


Figure 1: Layered NB-IoT architecture for smart utility monitoring

| Layer               | Component(s)                                                                                                           | Primary Function                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1. End Device       | Smart electricity/water/gas meters with embedded NB-IoT (Cat-NB1/NB2) modem, sensor front-end, local flash buffer      | Sense consumption/flow/pressure, timestamp readings, buffer locally, transmit on a scheduled/event basis                  |
| 2. Radio Access     | NB-IoT eNodeB (in-band, guard-band or standalone deployment); NPRACH, NPDCCH, NPDSCH, NPUSCH channels                  | Random access, scheduling, repetition-based link adaptation, coverage-class assignment, PSM/eDRX paging control           |
| 3. Core Network     | Evolved Packet Core: MME, S-GW/P-GW, HSS; SCEF with Non-IP Data Delivery (NIDD)                                        | Mobility/session management, authentication (EPS-AKA), lightweight small-data delivery bypassing the heavy IP/GTP stack   |
| 4. Service Platform | IoT device-management server, MQTT/CoAP broker, TLS/DTLS security gateway, protocol/data adapter                       | Device provisioning & lifecycle management, protocol translation, message queuing, secure API exposure to utility systems |
| 5. Application      | Meter Data Management System (MDMS), billing engine, GIS dashboard, AI analytics (load forecast, theft/leak detection) | Consumption aggregation & billing, outage/leak alerting, demand forecasting, regulatory reporting                         |

## 2.1 Coverage Enhancement Design

NB-IoT closes the deep-coverage gap of legacy cellular technologies by repeating the same transport block multiple times and combining the copies at the receiver, trading throughput and latency for link-budget gain. Devices are assigned one of three Coverage Enhancement (CE) classes based on measured signal quality, as summarised in Table 2.

| Coverage Class         | Target MCL | Typical Scenario                                   | Repetitions (approx.) |
|------------------------|------------|----------------------------------------------------|-----------------------|
| Normal Coverage (CE0)  | 144 dB     | Above-ground meters, line of sight to eNB          | 1                     |
| Robust Coverage (CE1)  | 154 dB     | Ground-floor / semi-shielded installations         | ~4–16                 |
| Extreme Coverage (CE2) | 164 dB     | Basements, underground pits, dense-urban shielding | ~128–2048             |

## 2.2 End-to-End Data Flow

- The meter senses consumption/flow, timestamps the reading and stores it in a local buffer.
- At a scheduled interval (e.g., daily) the modem wakes from Power-Saving Mode (PSM), performs NPRACH random access, and is assigned an appropriate coverage class and repetition count.
- The reading is transmitted uplink over NPUSCH (with repetitions if required) to the NB-IoT eNodeB.
- The eNodeB forwards the data through the EPC using Non-IP Data Delivery (NIDD) via the SCEF, avoiding the overhead of a full IP/PDN connection for a small payload.
- The IoT service platform authenticates, decodes and queues the message, exposing it through a secure API to the utility application layer.
- The MDMS/billing/analytics systems process the reading; any control message (e.g., an alert acknowledgement or a firmware flag) is queued and delivered on the device's next scheduled paging occasion (NPDCCH/NPDSCH) via eDRX.
- The device returns to PSM sleep until the next scheduled report or a paging event.

## 3. Evaluation of Design

Table 3 evaluates the proposed architecture against its own design objectives, contrasting the strengths that follow directly from the NB-IoT air interface with the residual gaps that a deployment team must plan around.

| Strengths                                                                                                        | Weaknesses / Design Gaps                                                                                                      |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Deep in-building/underground penetration (up to 164 dB MCL) via repetition — reaches basement/pit-mounted meters | High repetition counts increase latency (seconds to tens of seconds) — unsuitable for time-critical alarms such as a gas leak |
| Massive connection density (tens of thousands of devices per cell) suits city-scale AMI rollout                  | Repetition consumes more radio resource per device, reducing effective cell capacity under dense extreme-coverage load        |
| PSM/eDRX give multi-year battery life without truck-rolls for battery replacement                                | Very low throughput — cannot support firmware OTA at scale or high-rate telemetry                                             |
| Reuses licensed LTE spectrum/infrastructure via software upgrade — low incremental CAPEX, interference-protected | Single-operator/single-cell dependency — no defined fallback path if the serving cell is congested or unavailable             |
| 3GPP-grade security (EPS-AKA, ciphering) built into the access layer                                             | NIDD payloads are not inherently end-to-end encrypted — needs an explicit application-layer security addition                 |
| Low-cost, low-complexity modules suitable for mass utility deployment                                            | Static-device assumption — the architecture offers little benefit for mobile utility assets                                   |

The central design tension is that coverage, capacity and latency are coupled through the repetition mechanism: pushing MCL to 164 dB by increasing N necessarily lengthens transmission time and reduces the number of devices a cell can serve within a given reporting window. This is the principal gap the architecture must be evaluated against, since a smart-utility deployment mixes well-covered above-ground meters with a smaller population of extreme-coverage basement meters, and a single fixed configuration is unlikely to be optimal for both.

## 4. Justification

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The design choices above are justified quantitatively using standard link-budget and coverage-enhancement principles (validated in the MATLAB analysis of Section 6).

### 4.1 Link Budget and Maximum Coupling Loss

The uplink is the limiting direction for a battery-powered meter (23 dBm maximum transmit power). Using a thermal-noise-floor-based receiver sensitivity model with a 5 dB noise figure and a required SINR of about  $-12.6$  dB after processing, the baseline (single-shot) MCL for the assumed NB-IoT link works out to approximately 148 dB. This is short of the 164 dB Extreme Coverage target by about 16 dB, which is exactly the gap that repetition-based combining is designed to close ( $10 \cdot \log_{10}(N)$  dB of gain per doubling of N), justifying the inclusion of coverage-class-based repetition as a mandatory, not optional, element of the architecture.

### 4.2 Range Extension

Applying the resulting effective MCL to a COST-231 Hata dense-urban path-loss model at 900 MHz shows the achievable cell radius growing from roughly 3.5 km with no repetition to over 30 km at the highest repetition levels used for extreme coverage (see Section 6, Figure 2/Table 5). This confirms that a single NB-IoT cell, correctly configured with coverage classes, can serve both above-ground and deep-basement meters within one physical deployment, justifying the single-eNodeB-per-area design rather than a denser, more expensive cell layout.

### 4.3 Battery Life

Using PSM/eDRX duty cycling with one report per day, a few seconds of active transmit/receive current and microamp-level sleep current, the MATLAB battery model (Section 6) estimates an operational life well in excess of the ten-year design target for a typical primary-cell capacity, even under the worst-case Extreme Coverage transmit duration. This provides justified margin for real-world de-rating (temperature, self-discharge, occasional extra retransmissions) without violating the ten-year objective.

### 4.4 Core-Network Efficiency

Using NIDD/SCEF instead of a full IP/GTP bearer is justified because the payload per report (tens of bytes) is far smaller than the signalling overhead of establishing a conventional PDN connection; bypassing that overhead reduces both device energy consumption and core-network signalling load, which matters at the tens-of-thousands-of-devices-per-cell scale targeted by this design.

## 5. Suggestions for Improvement

The following enhancements address the design gaps identified in Section 3 while preserving the core NB-IoT architecture.

| Improvement                                                      | Rationale / Expected Benefit                                                                                                                                                      |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive repetition control (RSRP-triggered)                     | Continuously right-sizes the repetition count to the measured link instead of a fixed worst case, recovering capacity and reducing latency for well-covered meters                |
| Priority / expedited access for alarm traffic                    | A short, low-repetition emergency message class (e.g., gas-leak or tamper alert) bypasses the standard reporting queue to cut latency                                             |
| Local concentrator / mesh aggregation                            | A building- or street-level gateway aggregates several meters over a short-range link and relays over a single NB-IoT connection, cutting per-device signalling and cellular cost |
| Dual-connectivity / fallback bearer (LTE-M or 2G)                | Provides a backup path for firmware updates or when NB-IoT capacity/coverage is degraded, improving overall availability                                                          |
| End-to-end application-layer security (TLS/DTLS + PKI over NIDD) | Closes the gap between link-layer ciphering and true end-to-end confidentiality/integrity for billing-grade data                                                                  |
| Edge pre-processing / data compression on the meter              | Reduces payload size and transmission duration, directly lowering energy per report and improving battery-life margin                                                             |
| Multi-carrier NB-IoT and later-release (Rel-14+) features        | Additional carriers and TDD support increase aggregate cell capacity as device density grows over the network lifetime                                                            |

## 6. Supporting MATLAB Design Analysis

A MATLAB script (nbiot\_design\_analysis.m, provided in full below) was written to quantitatively validate the design and justification arguments above: (1) link budget / Maximum Coupling Loss, (2) coverage-enhancement SNR gain versus repetitions, (3) achievable cell range versus repetitions using a COST-231 Hata model, (4) battery-life estimation under PSM/eDRX, and (5) the throughput–coverage trade-off. Key output plots are shown below; a full results table is given in Table 5.

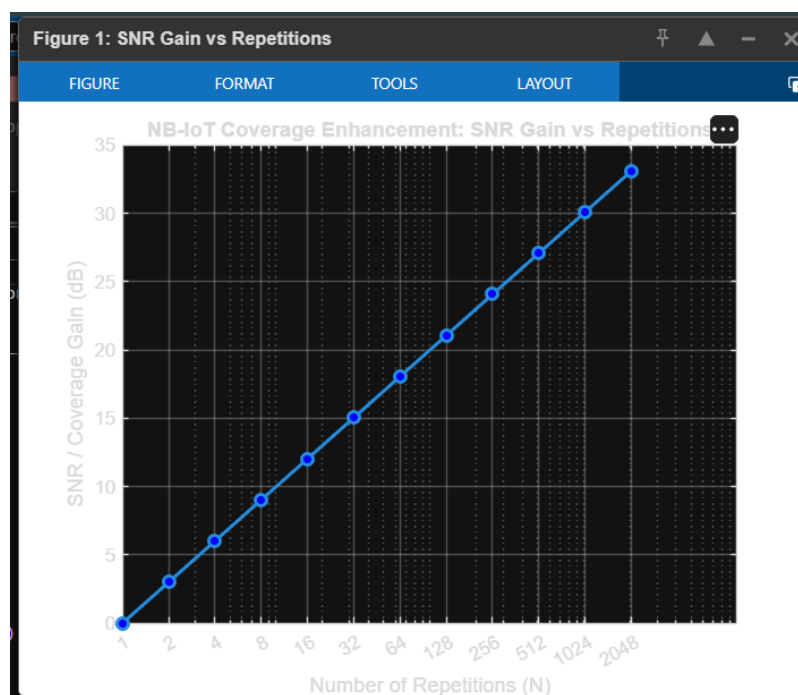


Figure 2: SNR / coverage gain versus number of repetitions



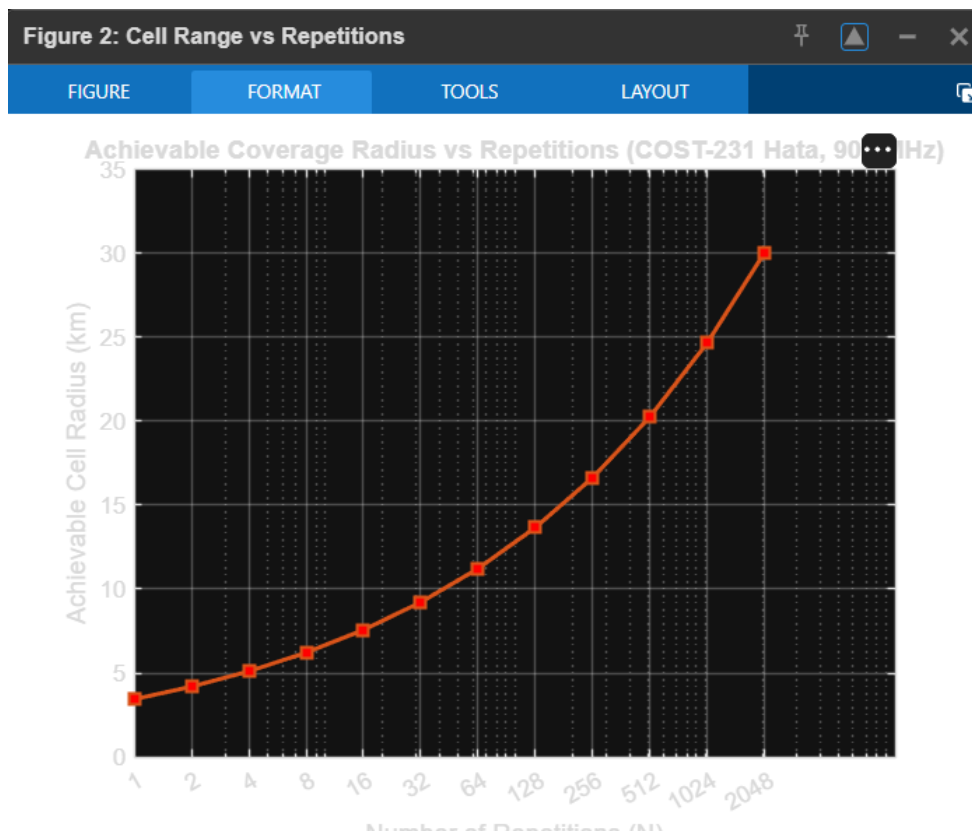


Figure 3: Achievable cell radius versus number of repetitions (COST-231 Hata, 900 MHz)

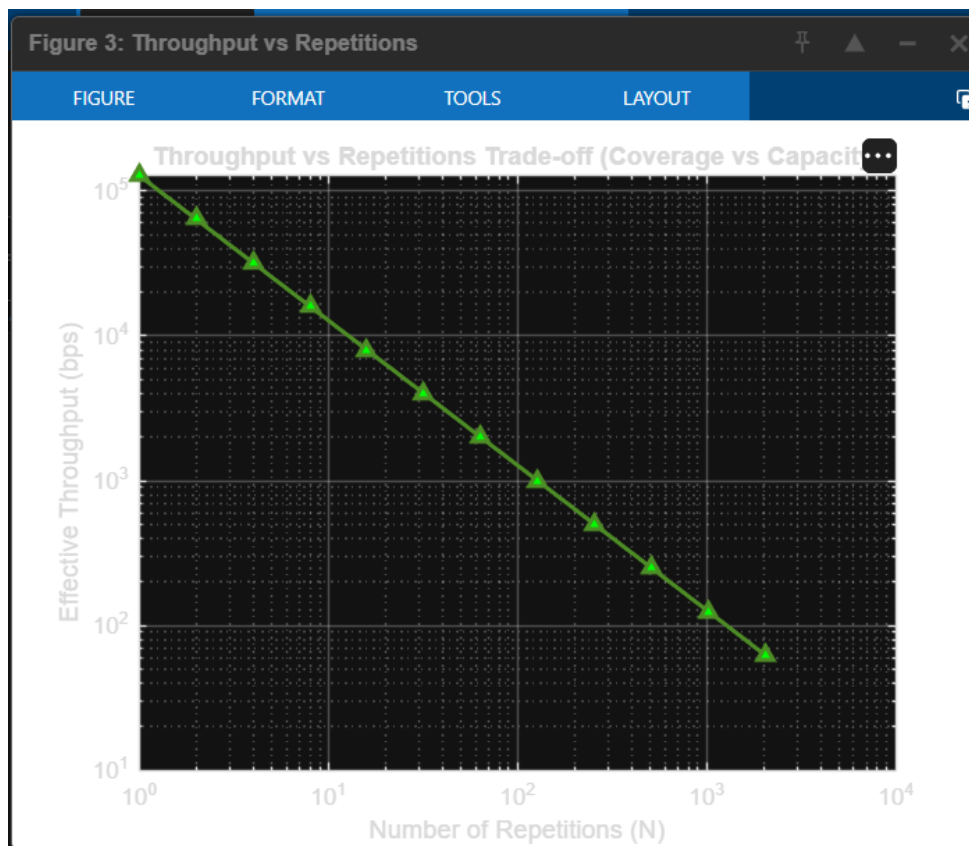


Figure 4: Effective throughput and range trade-off across repetition levels

| Repetitions (N) | SNR Gain (dB) | Achievable Range (km) | Effective Throughput (bps) |
|-----------------|---------------|-----------------------|----------------------------|
| 1               | 0.0           | 3.45                  | 128000                     |
| 16              | 12.0          | 7.57                  | 8000                       |
| 128             | 21.1          | 13.66                 | 1000                       |
| 512             | 27.1          | 20.25                 | 250                        |
| 2048            | 33.1          | 30.02                 | 62.5                       |

## MATLAB Source Code

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```
clc; clear; close all;

% Output folder for generated figures (created if it doesn't exist)
outDir = fullfile(pwd, 'nbiot_figures');
if ~exist(outDir, 'dir')
    mkdir(outDir);
end

%% -----
% SECTION 1: LINK BUDGET / MAXIMUM COUPLING LOSS (MCL)
% -----
% NB-IoT defines three Coverage Enhancement (CE) classes. MCL is computed as:
% MCL (dB) = Tx_Power + Tx_AntGain - RxSensitivity + Rx_AntGain - Margins
%
% Uplink is generally the limiting link for battery-powered meters (23 dBm UE).

Tx_Power_dBm      = 23;      % NB-IoT UE (Cat-NB1) max transmit power (dBm)
Rx_NoiseFloor_dBm = -173.9; % Thermal noise density (dBm/Hz) reference
BW_Hz             = 180e3;   % NB-IoT system bandwidth (Hz)
NF_dB             = 5;       % eNodeB receiver noise figure (dB)
ReqSINR_dB        = -12.6;   % Required SINR after repetition gain & processing
                        (dB)
InterferenceMargin_dB = 2;   % Rise-over-thermal / interference margin
BodyLoss_dB        = 0;      % Static utility meter -> negligible body loss
CableConnLoss_dB   = 2;      % Feeder/connector losses at eNodeB

RxSensitivity_dBm = Rx_NoiseFloor_dBm + 10*log10(BW_Hz) + NF_dB + ReqSINR_dB;

MCL_dB = Tx_Power_dBm - RxSensitivity_dBm - InterferenceMargin_dB -
CableConnLoss_dB - BodyLoss_dB;

fprintf('=====\\n');
fprintf(' SECTION 1: LINK BUDGET SUMMARY\\n');
fprintf('=====\\n');
fprintf('Receiver sensitivity           : %6.2f dBm\\n', RxSensitivity_dBm);
fprintf('Baseline Maximum Coupling Loss   : %6.2f dB\\n', MCL_dB);
```

```

% NB-IoT standard target MCL classes (3GPP TR 45.820 benchmark values)
CoverageClass = {'Normal Coverage (CE0)'; 'Robust Coverage (CE1)'; 'Extreme Coverage (CE2)'};
Target_MCL_dB = [144; 154; 164]; % 3GPP defined target MCL per class
RepetitionsRequired = ceil(10.^((Target_MCL_dB - MCL_dB)/10)); % repetitions needed
RepetitionsRequired = round(RepetitionsRequired); % guard against float precision
RepetitionsRequired(RepetitionsRequired < 1) = 1;

fprintf('%-26s %14s %18s\n','Coverage Class','Target MCL(dB)','Reps Required');
for k = 1:length(CoverageClass)
    fprintf('%-26s %14d %18d\n', CoverageClass{k}, Target_MCL_dB(k), RepetitionsRequired(k));
end

%% -----
% SECTION 2: COVERAGE-ENHANCEMENT SNR GAIN vs REPETITIONS
% -----
% Repeating the same transport block N times and combining at the receiver
% (chase combining) provides an approximate processing gain of  $10 \cdot \log_{10}(N)$ .

N_rep = [1 2 4 8 16 32 64 128 256 512 1024 2048];
SNR_Gain_dB = 10*log10(N_rep);

fig1 = figure('Name','SNR Gain vs Repetitions','Color','w');
semilogx(N_rep, SNR_Gain_dB, '-o', 'LineWidth', 1.8, 'MarkerFaceColor','b');
grid on; xlabel('Number of Repetitions (N)'); ylabel('SNR / Coverage Gain (dB)');
title('NB-IoT Coverage Enhancement: SNR Gain vs Repetitions');
xticks(N_rep);
saveas(fig1, fullfile(outDir, 'fig1_snr_gain_vs_repetitions.png'));

%% -----
% SECTION 3: ACHIEVABLE CELL RANGE vs REPETITIONS (COST-231 Hata Model)
% -----
% Urban macro-cell path-loss model (COST-231 Hata), typical for utility
% meters mounted at basements/street cabinets (dense urban deployment).

```

```

f_MHz    = 900;           % NB-IoT deployed in-band on 900 MHz LTE carrier
h_BS     = 30;            % eNodeB antenna height (m)
h_MS     = 1.5;           % Meter/module antenna height (m)
C         = 3;            % dense urban correction factor (dB)

a_hMS = (1.1*log10(f_MHz) - 0.7)*h_MS - (1.56*log10(f_MHz) - 0.8);

% Forward path-loss model, and its analytic inverse (both used/cross-checked
below)
PL_dB     = @(d_km) 46.3 + 33.9*log10(f_MHz) - 13.82*log10(h_BS) - a_hMS + ...
              (44.9 - 6.55*log10(h_BS)).*log10(d_km) + C;
PL_slope  = (44.9 - 6.55*log10(h_BS));
PL_offset = 46.3 + 33.9*log10(f_MHz) - 13.82*log10(h_BS) - a_hMS + C;
d_km_of   = @(PL) 10.^((PL - PL_offset) / PL_slope); % inverse of PL_dB

% Solve for distance d such that PL(d) = MCL_with_repetition
MCL_with_rep = MCL_dB + SNR_Gain_dB; % effective MCL after gain (dB)
d_km = d_km_of(MCL_with_rep);

% Cross-check: re-run forward model on solved distances, confirm round-trip
PL_check = PL_dB(d_km);
assert(max(abs(PL_check - MCL_with_rep)) < 1e-6, ...
    'Path-loss inversion check failed - review COST-231 Hata parameters.');
```

```

fig2 = figure('Name','Cell Range vs Repetitions','Color','w');
semilogx(N_rep, d_km, '-s', 'LineWidth', 1.8, 'Color', [0.85 0.33 0.10],
'MarkerFaceColor','r');
grid on; xlabel('Number of Repetitions (N)'); ylabel('Achievable Cell Radius
(km)');
title('Achievable Coverage Radius vs Repetitions (COST-231 Hata, 900 MHz)');
xticks(N_rep);
saveas(fig2, fullfile(outDir, 'fig2_range_vs_repetitions.png'));
```

```

fprintf('\n=====');
fprintf(' SECTION 3: RANGE EXTENSION SUMMARY\n');
fprintf('=====');
for k = 1:length(N_rep)
    fprintf('N = %4d reps | Gain = %5.1f dB | Range = %6.2f km\n', ...
```

```

        N_rep(k), SNR_Gain_dB(k), d_km(k));
end

%% -----
% SECTION 4: BATTERY-LIFE ESTIMATE (PSM / eDRX DUTY CYCLING)
% -----
% Smart meter reports consumption once per reporting interval; sleeps in
% Power-Saving-Mode (PSM) between reports. Battery life target: >=10 years.

BatteryCapacity_mAh = 5000;      % typical Li-SOCl2 primary cell used in meters
I_sleep_uA    = 3;               % deep-sleep (PSM) current (uA)
I_tx_mA       = 220;             % active transmit current during NPUSCH (mA)
I_idle_mA     = 6;               % idle/paging listening current (mA)

T_report_hours = 24;             % one report per day (typical AMI reporting)
T_tx_sec       = 3;              % active transmit duration per report incl.
repetitions (s) - worst case CE2
T_idle_sec     = 2;              % paging/idle listen duration per cycle (s)

Cycle_sec = T_report_hours*3600;
Charge_per_cycle_mAh = (I_tx_mA*T_tx_sec + I_idle_mA*T_idle_sec)/3600 + ...
(I_sleep_uA/1000)*(Cycle_sec - T_tx_sec -
T_idle_sec)/3600;

Cycles_per_year = 365*24*3600/Cycle_sec;
Annual_Consumption_mAh = Charge_per_cycle_mAh*Cycles_per_year;
BatteryLife_years = BatteryCapacity_mAh/Annual_Consumption_mAh;

fprintf('\n=====\\n');
fprintf(' SECTION 4: BATTERY LIFE ESTIMATE\\n');
fprintf('=====\\n');
fprintf('Charge consumed per reporting cycle : %8.5f mAh\\n',
Charge_per_cycle_mAh);
fprintf('Annual charge consumption           : %8.3f mAh/year\\n',
Annual_Consumption_mAh);
fprintf('Estimated battery life                 : %8.2f years\\n',
BatteryLife_years);

%% -----

```

```

% SECTION 5: THROUGHPUT vs REPETITION TRADE-OFF
% -----
% Higher repetitions extend coverage/reliability but reduce effective
% throughput and increase channel occupancy (capacity trade-off highlighted
% in the "Evaluation of Design" / "Justification" sections).

TB_bits = 256; % small transport block (typical meter
reading payload)
Tslot_ms = 2; % approx per-repetition unit time
(simplified)
Effective_Throughput_bps = TB_bits ./ (N_rep * Tslot_ms/1000);

fig3 = figure('Name','Throughput vs Repetitions','Color','w');
loglog(N_rep, Effective_Throughput_bps, '-^', 'LineWidth', 1.8, ...
    'Color', [0.30 0.55 0.15], 'MarkerFaceColor','g');
grid on; xlabel('Number of Repetitions (N)'); ylabel('Effective Throughput
(bps)');
title('Throughput vs Repetitions Trade-off (Coverage vs Capacity)');
saveas(fig3, fullfile(outDir, 'fig3_throughput_vs_repetitions.png'));

fprintf('\n=====');
fprintf(' SECTION 5: THROUGHPUT-COVERAGE TRADE-OFF\n');
fprintf('=====');
for k = 1:length(N_rep)
    fprintf('N = %4d | Throughput = %8.2f bps | Range = %6.2f km\n', ...
        N_rep(k), Effective_Throughput_bps(k), d_km(k));
end

fprintf('\nAnalysis complete. Figures saved to: %s\n', outDir);

```